Hints: Polynomials – 4. Real roots: Descartes, Sturm and Newton's inequalities

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Descartes: divide by a power of x so that $P(0) \neq 0$, and induct on the degree using Rolle: if P has Z positive roots, then P' has at least $Z-1$. How does the number of sign changes of P' compare with that of P?

Hint 2. The coefficients of P' are those of P (except the constant one) times positive numbers, so $V(P') \in \{V(P), V(P)-1\}$. Parity: the number of positive roots is even iff $P(0)$ and the leading coefficient have the same sign, iff $V(P)$ is even. Combine $Z(P) \leqslant Z(P')+1 \leqslant V(P')+1 \leqslant V(P)+1$ (induction) with $Z(P) \equiv V(P) \pmod 2$.

Hint 3. Newton: differentiate $P(x) = \sum \binom{n}{k} c_k x^k$ exactly $k-1$ times, reverse ($x^m Q(1/x)$), and differentiate again until a quadratic is left. Rolle and reversal preserve real-rootedness, and the quadratic is a positive multiple of $c_{k-1} + 2c_k x + c_{k+1}x^2$; its discriminant gives $c_k^2 \geqslant c_{k-1}c_{k+1}$.

Hint 4. The binomial coefficients are log-concave, so $a_k = \binom{n}{k} c_k$ satisfies $a_k^2 \geqslant a_{k-1}a_{k+1}$ too. Example: $x^2 + x + 1$ has log-concave coefficients $1, 1, 1$ but no real roots.

Problem 2

Hint 1. The roots, if real, are negative. Try to find $n+1$ negative points where the polynomial alternates in sign. Natural candidates: points where one term dominates.

Hint 2. At $x = -2^{2j-n}$ the k-th term equals $(-1)^k 2^{j^2-(k-j)^2}$. So the term $k = j$ has size 2^{j^2}, the two neighbours $k = j \pm 1$ each have size 2^{j^2-1}, and so on.

Hint 3. The value is $(-1)^j 2^{j^2} \sum_d (-1)^d 2^{-d^2}$, summed over the admissible $d = k - j$. Group the terms $d = 0$, $d = \pm 1$ (total $\geqslant 0$), then $d = \pm 2, \pm 3, \ldots$ as an alternating series with rapidly decreasing terms, to show the bracket is positive. So the signs alternate at $j = 0, \ldots, n$ and there are n real roots.

Problem 3

Hint 1. Relate $P_{n,p}$ to the n-th derivative of $t^p(1-t)^p$, which by Rolle has n roots in $(0,1)$ when $n \leqslant p$.

Hint 2. By Leibniz, $\frac{d^n}{dt^n}\left[t^p(1-t)^p\right] = \sum_j \binom{n}{j} \frac{p!}{(p-j)!} t^{p-j} \cdot (-1)^{n-j} \frac{p!}{(p-n+j)!} (1-t)^{p-n+j}$. Factor out $n!\, t^p (1-t)^{p-n}$.

Hint 3. You get $(-1)^n n!\, t^p (1-t)^{p-n}\, P_{n,p}\!\left(-\frac{1-t}{t}\right)$. Repeated Rolle (the multiplicities at 0 and 1 stay positive up to the n-th derivative) gives n distinct roots in $(0,1)$, and $t \mapsto -\frac{1-t}{t}$ maps $(0,1)$ bijectively onto $(-\infty, 0)$.

Problem 4

Hint 1. Taylor's formula for polynomials: $P(x+h) = \sum_k \frac{P^{(k)}(x)}{k!} h^k$. Which choice of h (or combination) produces only the even derivatives?

Hint 2. $Q(x) = \frac{1}{2}\big(P(x+1) + P(x-1)\big)$. A real polynomial without real roots has constant sign, so Q has the same constant sign.

Problem 5

(a)

Hint 1. Look at $\frac{P'}{P} = \sum_i \frac{1}{x - x_i}$ and its derivative $\frac{PP'' - P'^2}{P^2} = -\sum_i \frac{1}{(x-x_i)^2}$.

Hint 2. For x not a root, Cauchy–Schwarz gives $\left(\sum \frac{1}{x-x_i}\right)^2 \leqslant n \sum \frac{1}{(x-x_i)^2}$, i.e. $P'^2 \leqslant n(P'^2 - PP'')$, which is the claim. At a root of P the claim reads $(n-1)P'(x)^2 \geqslant 0$.

(b)

Hint 1. When is Cauchy–Schwarz an equality? Treat roots of P separately.

Hint 2. At a non-root x: equality iff all $x - x_i$ are equal, i.e. $P = c(x-a)^n$ (and then equality holds for all x). At a root x: equality iff $n = 1$ or x is a multiple root.